

Riddim Protocol — Product Requirements Document

Event: EAG Global Buildathon — Nigeria **Tracks:** AI-Native Creator Economy & Digital Rights (primary) · Real-World Ethereum Applications (secondary) · light touch on AI x Ethereum & Agent Economy **Build window:** 9:30am–1:30pm, Aug 27 (submission closes 2:00pm)

1. Problem

Nigerian and African music has no functioning mechanical-licensing infrastructure. Collection societies (COSON, MCSN) are widely regarded as dysfunctional, so producers and session contributors have no reliable way to get paid when their work is reused.

This is worsened by two specific, current phenomena:

Riddim/beat reuse is standard practice in Afrobeats and Amapiano — a single instrumental is licensed and reused across many different songs by different artists — but the original producer is routinely under-credited and underpaid when the derivative blows up.

AI voice cloning of artists is now widespread — AI-generated covers/remixes using cloned artist voices circulate and monetize with zero consent and zero payment to the artist.

Both are the same underlying failure: **no programmable link between "who contributed what" and "who gets paid when it earns money."**

2. Solution

Riddim Protocol is an onchain registry where:

A producer registers a riddim as a set of **components** (e.g. drums, melody, vocal hook), each with a preset ownership split and payout wallet.

An artist "licenses" that riddim to build a new track — this is a simple onchain call, not a legal negotiation.

Any revenue the new track earns (modeled as stablecoin tips for the demo) runs through a **split function** that automatically pays every component owner their percentage — cascading back to the original producer.

AI voice-clones are treated as just another component type — an artist can register consent + a royalty rate for their cloned voice being used in AI covers, using the exact same registry and split logic.

A **stubbed AI auto-detect** feature demonstrates the intended end-state: an audio-similarity check that flags when a new upload likely reuses a registered riddim/voice, and suggests (does not force) the licensing step. Not fully functional — a clearly-labeled proof-of-concept, not production detection.

3. Target users (for the pitch, not full personas)

Producers/beatmakers — want to be paid automatically when their riddim is reused, without chasing labels or collection societies.

Artists/remixers — want a fast, transparent way to license a riddim or a voice without legal friction.

Fans — want a simple way to directly support the people behind a track they like.

4. MVP scope (in for the hackathon)

Register a riddim with 2–3 components and preset splits (onchain).

License a riddim to a new "track" record (onchain).

Register a voice-clone consent license using the same component mechanism.

Tip a track in a test stablecoin → contract auto-splits to all component owners.

A minimal frontend: register view, browse/license view, tip view, with the split visible on a block explorer link.

AI auto-detect stub: given two audio inputs (or precomputed mock fingerprints), return a similarity score and, above a threshold, pre-fill the "license this riddim" action for the user to confirm. Explicitly presented as a proof-of-concept, not real audio fingerprinting.

5. Explicitly out of scope (say this out loud in the pitch — it builds credibility, not weakness)

Real audio fingerprinting/MIR (Shazam-style matching) — stubbed only.

Legal enforceability of onchain licenses — flagged as future work, same as NUSIC and other prior art in this space.

Mainnet deployment — testnet only for the demo.

Full royalty accounting across real streaming platforms — out of scope; demo uses direct tips only.

Governance/dispute resolution for contested ownership splits.

6. Demo narrative (what judges see in 3 minutes)

Open with the real problem: AI covers of Nigerian artists' voices going viral with zero consent or pay, and riddim reuse leaving producers uncredited.

Show a producer registering a riddim with a split.

Show an artist licensing it into a new track.

Show the AI auto-detect stub flagging a similarity match and suggesting the license step.

Show a fan tipping the track — split executes live, visible on-chain.

Close with roadmap: real fingerprinting, artist onboarding, partnerships with African platforms like Boomplay/Audiomack.

7. Success criteria for the hackathon

Split payment executes correctly and visibly onchain during the live demo — this matters more than any other single thing.

Judges can clearly map the project to the stated tracks without you over-explaining.

The AI auto-detect piece is presented honestly as a proof-of-concept touchpoint, not oversold as functional AI.

8. Roadmap beyond the hackathon (for the "future plans" section of submission)

Real audio fingerprinting integration for automatic riddim/voice-clone detection.

Artist and producer onboarding flow with identity verification.

Mainnet deployment and integration with real streaming/tipping platforms.

Exploration of a lightweight dispute-resolution or attestation mechanism for contested splits.